

## GREENVILLE-SPARTANBURG, SC, USA

WMO: 723120

Lat: 34.906N

Lon: 82.213W

Elev: 955

StdP: 14.20

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03870

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 21.6       | 25.4 | 2.3                           | 6.4 | 30.3 | 6.5 | 8.0 | 32.7 | 22.6                  | 49.8 | 19.9 | 49.7 | 5.1                   | 340  | 0.406 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.5                   | 94.1            | 73.4 | 91.8 | 73.2 | 89.5 | 72.7 | 77.0                | 87.5 | 75.9 | 85.9 | 75.1 | 84.5 | 7.2                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.1                            | 131.8 | 80.0 | 73.2 | 127.7 | 79.2 | 72.3 | 124.0 | 78.6 | 40.8          | 87.4 | 39.9 | 86.3 | 39.0 | 84.9 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |      | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |     |      |
|-------------------|------|------|----------------------------|------|--------------------|------|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-----|------|
|                   |      |      | Mean                       |      | Standard Deviation |      | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |     |      |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max  | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |     |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h)  | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |     |      |
| 18.9              | 16.9 | 15.0 | 14.5                       | 97.4 | 4.2                | 3.8  | 11.5                                               | 100.1 | 9.0        | 102.3 | 6.7        | 104.4 | 3.6        | 107.1 |     |      |
|                   |      |      | DB                         | WB   | 12.1               | 78.7 | 3.6                                                | 1.6   | 9.5        | 79.9  | 7.4        | 80.8  | 5.3        | 81.8  | 2.7 | 82.9 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 61.6   | 42.5  | 46.2  | 53.0  | 61.4  | 69.2  | 76.5  | 79.4  | 78.4  | 72.7  | 62.1  | 51.9  | 45.3 |
|                                                                               | DBStd       | 14.84  | 9.51  | 8.78  | 8.85  | 7.30  | 6.15  | 4.55  | 3.69  | 3.89  | 5.52  | 7.30  | 7.90  | 8.57 |
|                                                                               | HDD50       | 787    | 271   | 163   | 72    | 7     | 0     | 0     | 0     | 0     | 0     | 5     | 71    | 198  |
|                                                                               | HDD65       | 2950   | 698   | 528   | 379   | 151   | 34    | 2     | 0     | 0     | 9     | 141   | 396   | 612  |
|                                                                               | CDD50       | 5031   | 38    | 56    | 167   | 349   | 595   | 794   | 910   | 881   | 680   | 380   | 128   | 53   |
|                                                                               | CDD65       | 1718   | 0     | 1     | 9     | 43    | 164   | 346   | 445   | 416   | 239   | 51    | 3     | 1    |
|                                                                               | CDH74       | 14610  | 1     | 6     | 85    | 426   | 1340  | 3022  | 4075  | 3569  | 1718  | 345   | 21    | 2    |
|                                                                               | CDH80       | 5703   | 0     | 0     | 10    | 76    | 393   | 1248  | 1817  | 1494  | 601   | 64    | 0     | 0    |
| Wind                                                                          | WSAvg       | 6.2    | 6.9   | 7.2   | 7.6   | 7.6   | 6.5   | 5.8   | 5.3   | 4.9   | 5.5   | 5.6   | 5.8   | 6.1  |
| Precipitation                                                                 | PrecAvg     | 50.2   | 4.1   | 4.2   | 5.2   | 3.7   | 4.1   | 4.5   | 4.7   | 4.3   | 3.9   | 3.8   | 3.6   | 4.0  |
|                                                                               | PrecMax     | 70.4   | 7.2   | 8.3   | 11.4  | 11.3  | 8.9   | 10.1  | 13.6  | 17.4  | 11.7  | 10.2  | 7.8   | 10.1 |
|                                                                               | PrecMin     | 35.0   | 0.3   | 0.6   | 1.1   | 0.7   | 1.1   | 0.2   | 0.7   | 0.8   | 0.2   | 0.0   | 0.6   | 0.4  |
|                                                                               | PrecStd     | 8.7    | 1.7   | 2.0   | 2.5   | 2.4   | 2.1   | 2.4   | 2.7   | 3.1   | 2.8   | 2.6   | 1.5   | 2.0  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 70.2  | 74.1  | 81.2  | 85.5  | 91.0  | 95.5  | 97.0  | 98.0  | 92.7  | 85.3  | 77.2  | 72.0 |
|                                                                               |             | MCWB   | 59.1  | 60.5  | 62.1  | 65.6  | 70.4  | 73.1  | 74.4  | 73.6  | 70.2  | 69.2  | 62.1  | 61.1 |
|                                                                               | 2%          | DB     | 65.4  | 68.9  | 76.1  | 82.2  | 87.3  | 92.5  | 94.0  | 93.5  | 89.6  | 81.2  | 72.6  | 67.2 |
|                                                                               |             | MCWB   | 57.1  | 56.1  | 59.7  | 64.1  | 69.2  | 72.6  | 74.4  | 73.5  | 70.6  | 66.3  | 60.0  | 59.6 |
|                                                                               | 5%          | DB     | 61.7  | 64.9  | 72.3  | 79.2  | 84.5  | 90.0  | 91.6  | 90.4  | 86.5  | 78.0  | 69.0  | 63.1 |
|                                                                               |             | MCWB   | 54.6  | 54.9  | 57.5  | 62.5  | 68.8  | 72.3  | 74.2  | 73.6  | 70.4  | 65.0  | 59.1  | 56.8 |
|                                                                               | 10%         | DB     | 57.3  | 60.9  | 68.2  | 75.7  | 81.6  | 87.1  | 89.1  | 87.7  | 83.4  | 74.3  | 65.4  | 59.1 |
|                                                                               |             | MCWB   | 51.1  | 51.7  | 55.4  | 60.8  | 67.3  | 71.4  | 73.7  | 73.2  | 69.6  | 63.8  | 57.4  | 53.0 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 63.4  | 64.2  | 65.2  | 69.3  | 74.2  | 76.9  | 78.2  | 78.2  | 75.4  | 73.8  | 68.7  | 65.8 |
|                                                                               |             | MCDB   | 66.9  | 69.9  | 73.2  | 80.1  | 83.8  | 88.3  | 89.9  | 89.0  | 84.1  | 79.5  | 70.8  | 68.2 |
|                                                                               | 2%          | WB     | 60.2  | 61.3  | 62.8  | 67.4  | 72.3  | 75.4  | 77.1  | 77.0  | 74.3  | 70.9  | 64.5  | 61.9 |
|                                                                               |             | MCDB   | 63.7  | 66.2  | 71.3  | 77.4  | 82.0  | 86.5  | 88.3  | 86.8  | 82.1  | 76.5  | 68.3  | 65.3 |
|                                                                               | 5%          | WB     | 56.3  | 57.6  | 60.6  | 65.4  | 70.9  | 74.4  | 76.0  | 75.8  | 73.0  | 68.5  | 62.1  | 58.5 |
|                                                                               |             | MCDB   | 60.3  | 62.5  | 68.5  | 73.8  | 80.1  | 84.9  | 86.6  | 84.9  | 80.3  | 74.1  | 66.3  | 62.2 |
|                                                                               | 10%         | WB     | 52.1  | 53.6  | 58.4  | 63.6  | 69.4  | 73.3  | 75.1  | 74.9  | 71.8  | 66.2  | 59.2  | 54.1 |
|                                                                               |             | MCDB   | 55.5  | 58.2  | 65.4  | 71.7  | 78.2  | 83.0  | 85.0  | 83.4  | 78.9  | 71.7  | 64.0  | 57.3 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 18.4  | 19.5  | 20.9  | 21.7  | 20.3  | 19.2  | 18.5  | 17.8  | 18.0  | 19.8  | 20.0  | 17.5 |
|                                                                               |             | MCDBR  | 22.2  | 23.8  | 25.4  | 25.3  | 23.3  | 22.8  | 21.9  | 21.6  | 22.0  | 22.9  | 23.9  | 20.7 |
|                                                                               |             | MCWBR  | 14.9  | 15.1  | 13.6  | 11.3  | 9.3   | 7.5   | 6.7   | 6.6   | 7.6   | 10.4  | 13.4  | 13.6 |
|                                                                               | 5% WB       | MCDBR  | 17.9  | 19.5  | 21.4  | 20.8  | 19.5  | 19.4  | 18.9  | 17.7  | 16.9  | 17.2  | 17.5  | 16.9 |
|                                                                               |             | MCWBR  | 14.2  | 14.9  | 13.5  | 10.9  | 8.7   | 7.5   | 6.9   | 6.6   | 7.1   | 9.8   | 13.3  | 13.4 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.310  | 0.320 | 0.354 | 0.394 | 0.452 | 0.479 | 0.525 | 0.513 | 0.416 | 0.379 | 0.333 | 0.308 |      |
|                                                                               | taud        | 2.509  | 2.479 | 2.412 | 2.298 | 2.172 | 2.133 | 2.030 | 2.047 | 2.299 | 2.385 | 2.473 | 2.545 |      |
|                                                                               | Ebn at Noon | 285    | 294   | 291   | 283   | 267   | 258   | 246   | 246   | 267   | 269   | 273   | 279   |      |
|                                                                               | Edh at Noon | 26     | 30    | 35    | 41    | 47    | 49    | 54    | 52    | 39    | 33    | 27    | 24    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 844    | 1045  | 1383  | 1723  | 1905  | 2009  | 1881  | 1738  | 1493  | 1242  | 936   | 725   |      |
|                                                                               | RadStd      | 48     | 117   | 93    | 127   | 148   | 142   | 123   | 83    | 133   | 141   | 83    | 69    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | +0.92 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +252  | +180  |
| Regional (54 neighbors) | +1.01 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +306  | +203  |

Nomenclature: See separate page